Language-Switching Invariance of LLM-as-a-Judge for Hindi Scripts and Hinglish — Step 4 run and results

Data: LLMBar (419 pairwise items) → 387-item human-validated core, rendered in English, Hindi (Devanagari), Romanized Hindi and code-mixed Hinglish. Six judges, two answer orders, temperature 0, Cloudflare AI Gateway (cache off).

> **Main findings.**
>
> 1. All six judges were less accurate in every Hindi-language condition than in English, and each of the 18 judge × language drops is significant after Holm correction (paired McNemar, $p_{Holm} \le 0.02$).
> 2. Pooled accuracy: English 78.9%, Devanagari 74.1%, Hinglish 73.3%, Romanized 70.3%. **Romanized Hindi is the hardest condition**: an item-clustered GEE gives odds ratio 0.60 [0.54, 0.67] for a correct verdict, compared with English.
> 3. Judges flip their verdict on 8–25% of identical item/order pairs when only the language/script changes. That is well above their own same-prompt re-run flip rate (0–11%).
> 4. Judges are affected differently (variant × judge interaction Wald χ^2=26.0, df=15, p=0.038). GPT-OSS-120B is the most robust (−3 pp); Llama-3.3-70B and Claude Haiku 4.5 lose 11–13 pp on Romanized Hindi.
> 5. The effect remains when the 12 code-heavy items or the 34 human-fixed items are removed (Table 4).

1. Step 4 run completeness

Run	Design	Expected	Present	Valid
Main (run 0)	387 items × 4 languages × 2 orders × 6 judges	18,576	18,576	18,570
Repeat 1, 2	100 fixed items (seed 2027) × 4 × 2 × 6, twice	9,600	9,600	9,600

The 6 unreadable judgments remained unreadable after 3 attempts each: 3 Qwen3 outputs where reasoning was cut off at the token limit, and 3 Llama-3.1-8B refusals. They are kept in the logs and **counted as incorrect** (a conservative choice). The run made 32,010 API calls, including retries of HTTP 429 rate-limit errors, at an estimated cost of $11.53 (token-based estimate). Every record stores the prompt hash, returned model, translation version, timestamp, token usage and raw response.

2. Method (algorithm)

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Algorithm 1: Controlled language-switching meta-evaluation
Input: items D = {(x, y1, y2, g)}, languages L = {en, deva, roman, cm}, judges J, orders O = {AB, BA}
1  for each item d in D, language l in L:  (x, y1, y2)_l ← validated translation T_l(d)   # human-
checked
2  for each judge j, item d, language l, order o:
3      prompt ← fixed LLMBar judge template filled with (x_l, first, second) per o
4      r ← j(prompt; temperature=0, no cache);   parse last "Output (a|b)" → pick p ∈ {1,2}
5      if unparseable or API error: retry (≤3 attempts) else store (p, raw, usage, hashes)
6  repeat lines 2–5 for runs k = 1, 2 on a fixed 100-item subset (noise floor)
7  compute accuracy, flip rate, position consistency per (j, l); pair cells on (d, o)
8  test l vs en: exact McNemar per judge (Holm across 18 tests); item-clustered bootstrap CIs (B=2000)
9  pooled model: GEE logistic correct ~ language + judge + order, clustered by item
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3. Metrics and equations

For judge j, language ℓ, item i, order o: the pick is $p_{j\ell io} \in \{1, 2\}$ and the gold label is g_i.

Accuracy:

$$\mathrm{Acc}_{j\ell} = \frac{1}{2|D|} \sum_{i \in D} \sum_{o \in \{AB, BA\}} \mathbb{1}[p_{j\ell io} = g_i]$$

Language-switch flip rate (relative to English):

$$\mathrm{Flip}_{j\ell} = \frac{1}{2|D|} \sum_{i,o} \mathbb{1}[p_{j\ell io} \neq p_{j,\mathrm{en},io}]$$

Accuracy gap: $\Delta_{j\ell} = \mathrm{Acc}_{j\ell} - \mathrm{Acc}_{j,\mathrm{en}}$. Position consistency: $\mathrm{PC}_{j\ell} = \frac{1}{|D|} \sum_i \mathbb{1}[p_{j\ell i,AB} = p_{j\ell i,BA}]$.

Re-run noise floor: across runs $k < k'$ on the subset D_{100}, $\mathrm{SelfFlip}_{j\ell} = \binom{3}{2}^{-1} \sum_{k<k'} \frac{1}{2|D_{100}|} \sum_{i,o} \mathbb{1}[p^{(k)}_{j\ell io} \neq p^{(k')}_{j\ell io}]$, and $\mathrm{Excess}_{j\ell} = \mathrm{Flip}_{j\ell} - \mathrm{SelfFlip}_{j\ell}$.

Exact McNemar test on the paired cells, with b = right in English and wrong in ℓ, c = the reverse:

$$p = P\big(X \leq \min(b, c)\big) \cdot 2, \quad X \sim \mathrm{Bin}(b + c, \tfrac{1}{2})$$

GEE logistic model with exchangeable within-item correlation:

$$\mathrm{logit}\, P(y_{ij\ell o} = 1) = \beta_0 + \beta_\ell + \gamma_j + \delta_o, \qquad \mathrm{OR}_\ell = e^{\beta_\ell}$$

Confidence intervals come from 2000 bootstrap resamples of items (the clusters), so the two orders and the paired cells of an item are resampled together.

4. Results

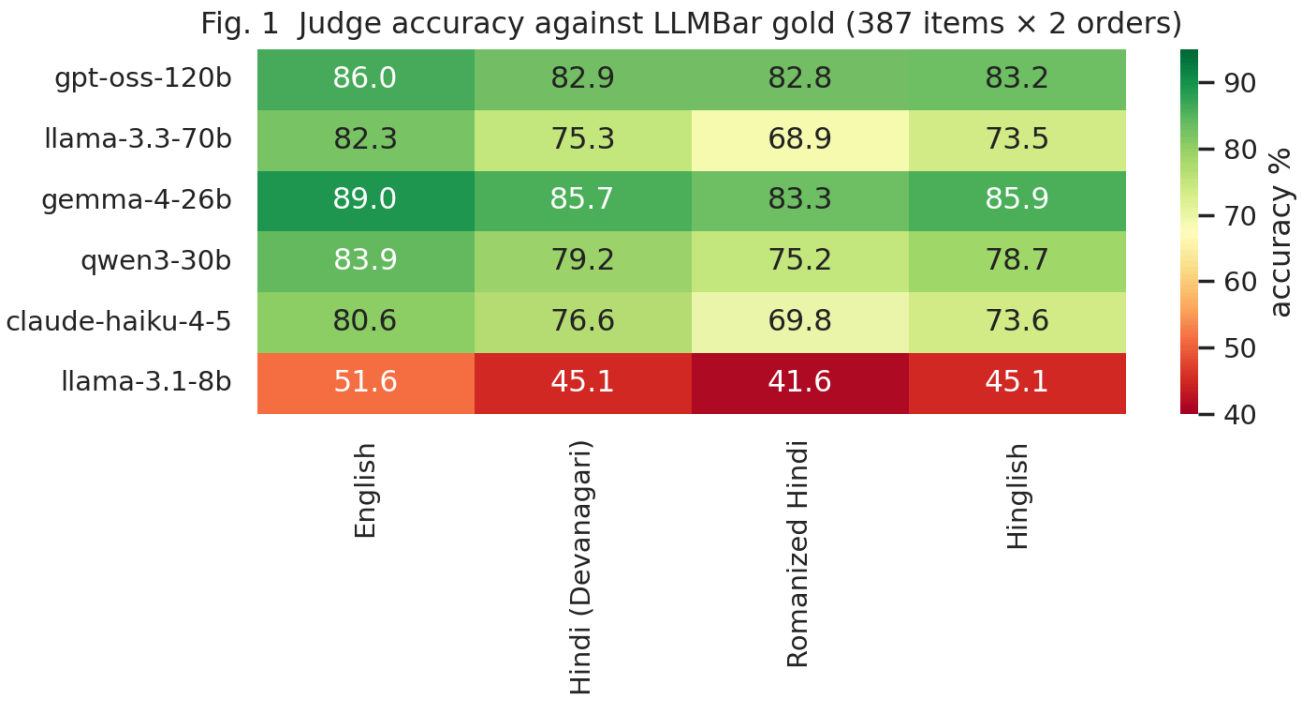


Figure 1. Accuracy (%) of each judge in each language condition.

Table 1. Per-judge results. Δacc and flip are relative to English; 95% item-clustered bootstrap CIs in brackets; p is Holm-adjusted McNemar.

Judge	Language	Acc %	Δacc pp [95% CI]	Flip % [95% CI]	Pos. consistency %	First-pos %	p (Holm)
gpt-oss-120b	**English**	**86.0**	—	—	**94.3**	**52.3**	—
gpt-oss-120b	Hindi (Devanagari)	82.9	-3.1 [-5.6, -0.5]	8.8 [6.6, 11.2]	91.7	52.3	0.02
gpt-oss-120b	Romanized Hindi	82.8	-3.2 [-5.9, -0.5]	10.5 [7.9, 13.0]	91.5	50.9	0.02
gpt-oss-120b	Hinglish	83.2	-2.8 [-5.4, -0.5]	8.5 [6.3, 11.0]	91.2	52.3	0.02
llama-3.3-70b	**English**	**82.3**	—	—	**88.9**	**52.7**	—
llama-3.3-70b	Hindi (Devanagari)	75.3	-7.0 [-10.2, -3.9]	16.3 [13.6, 19.1]	79.6	57.1	1.8e-05
llama-3.3-70b	Romanized Hindi	68.9	-13.4 [-16.9, -9.8]	21.2 [17.8, 24.5]	77.0	59.4	1.2e-15
llama-3.3-70b	Hinglish	73.5	-8.8 [-12.1, -5.7]	16.3 [13.4, 19.3]	78.0	58.1	1.5e-08
gemma-4-26b	**English**	**89.0**	—	—	**91.5**	**49.6**	—
gemma-4-26b	Hindi (Devanagari)	85.7	-3.4 [-5.6, -1.2]	9.3 [7.2, 11.6]	89.9	50.4	0.018
gemma-4-26b	Romanized Hindi	83.3	-5.7 [-8.3, -3.2]	10.6 [8.1, 13.2]	89.9	51.0	1.4e-05
gemma-4-26b	Hinglish	85.9	-3.1 [-5.2, -1.2]	8.3 [6.1, 10.6]	91.5	49.4	0.018
qwen3-30b	**English**	**83.9**	—	—	**87.6**	**56.1**	—
qwen3-30b	Hindi (Devanagari)	79.2	-4.7 [-7.6, -1.8]	13.8 [11.2, 16.8]	84.8	53.2	0.0048
qwen3-30b	Romanized Hindi	75.2	-8.7 [-12.0, -5.4]	17.1 [14.2, 20.3]	83.7	55.8	7.4e-08
qwen3-30b	Hinglish	78.7	-5.2 [-8.0, -2.2]	13.2 [10.6, 16.0]	84.7	56.7	0.00099
claude-haiku-4-5	**English**	**80.6**	—	—	**87.6**	**53.4**	—
claude-haiku-4-5	Hindi (Devanagari)	76.6	-4.0 [-7.0, -1.0]	15.4 [12.5, 18.3]	83.2	53.5	0.02
claude-haiku-4-5	Romanized Hindi	69.8	-10.9 [-14.5, -7.0]	21.2 [18.0, 24.4]	79.8	53.9	5.7e-10
claude-haiku-4-5	Hinglish	73.6	-7.0 [-10.3, -3.7]	15.2 [12.4, 18.1]	84.0	54.9	9e-06
llama-3.1-8b	**English**	**51.6**	—	—	**68.6**	**61.6**	—
llama-3.1-8b	Hindi (Devanagari)	45.1	-6.5 [-9.8, -3.0]	21.8 [18.8, 24.7]	66.1	64.3	0.0011
llama-3.1-8b	Romanized Hindi	41.6	-9.9 [-13.8, -6.2]	25.3 [22.0, 28.6]	65.9	63.4	4.9e-07
llama-3.1-8b	Hinglish	45.1	-6.5 [-9.7, -3.4]	18.4 [15.8, 21.1]	67.2	63.3	0.00033

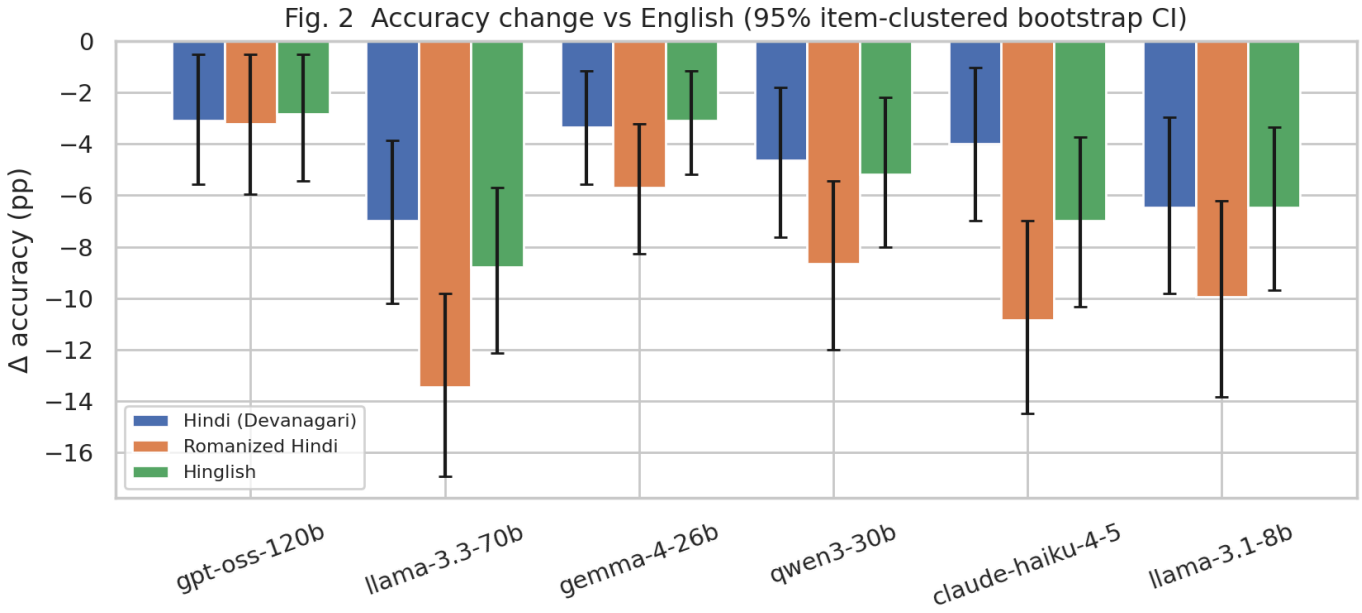


Figure 2. Accuracy change vs English with 95% CIs. All 18 bars exclude 0.

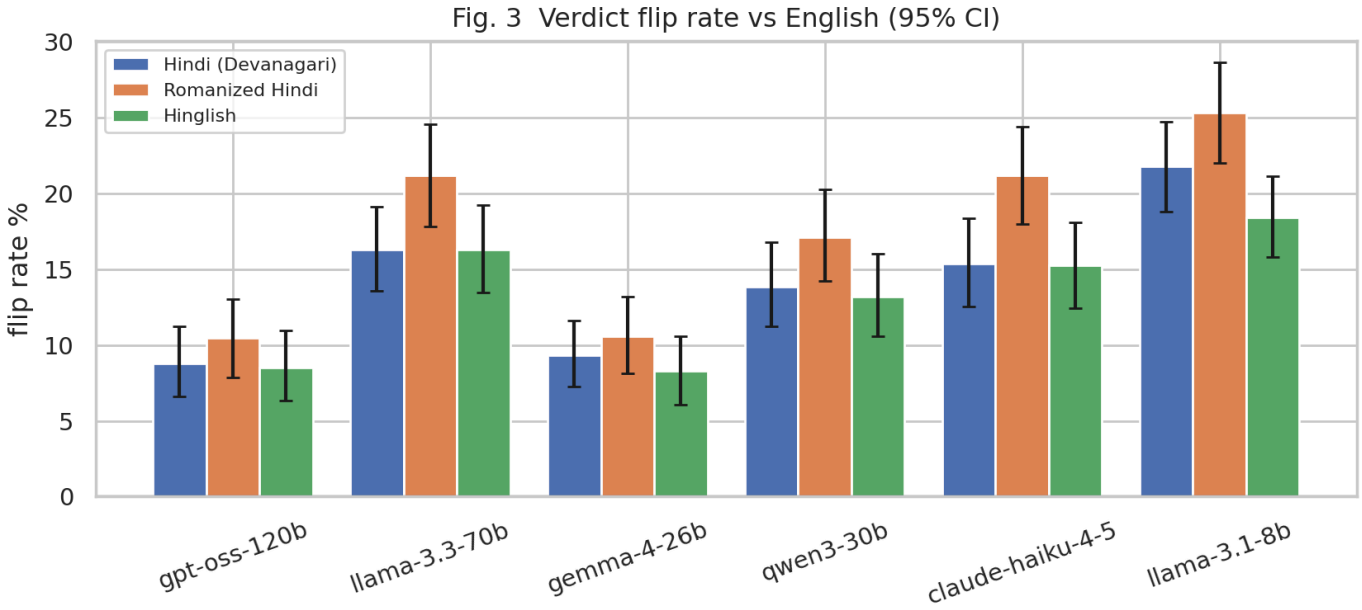


Figure 3. Fraction of identical item/order pairs whose verdict changes when only the language changes.

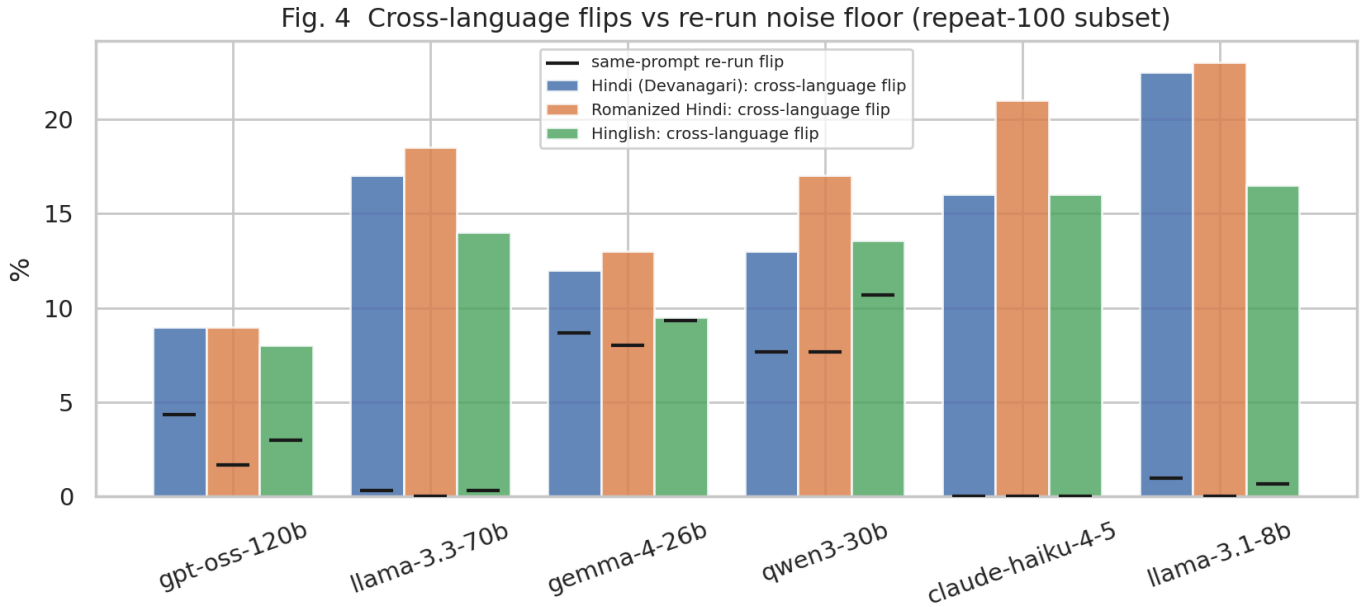


Figure 4. Cross-language flips (bars) vs the same-prompt re-run flip rate (black ticks) on the 100-item repeat subset.

Table 2. Repeat-run stability on the 100-item subset (3 runs).

Judge	Language	Self-flip %	All 3 runs identical %	Cross-language flip % (subset)	Excess pp
gpt-oss-120b	English	1.7	97.5	—	—
gpt-oss-120b	Hindi (Devanagari)	4.3	93.5	9.0	+4.7
gpt-oss-120b	Romanized Hindi	1.7	97.5	9.0	+7.3
gpt-oss-120b	Hinglish	3.0	95.5	8.0	+5.0
llama-3.3-70b	English	0.0	100.0	—	—
llama-3.3-70b	Hindi (Devanagari)	0.3	99.5	17.0	+16.7
llama-3.3-70b	Romanized Hindi	0.0	100.0	18.5	+18.5
llama-3.3-70b	Hinglish	0.3	99.5	14.0	+13.7

gemma-4-26b	English	7.0	89.5	—	—
gemma-4-26b	Hindi (Devanagari)	8.7	87.0	12.0	+3.3
gemma-4-26b	Romanized Hindi	8.0	88.0	13.0	+5.0
gemma-4-26b	Hinglish	9.3	86.0	9.5	+0.2
qwen3-30b	English	7.3	89.0	—	—
qwen3-30b	Hindi (Devanagari)	7.7	88.5	13.0	+5.3
qwen3-30b	Romanized Hindi	7.7	88.5	17.0	+9.3
qwen3-30b	Hinglish	10.7	84.0	13.6	+2.9
claude-haiku-4-5	English	0.7	99.0	—	—
claude-haiku-4-5	Hindi (Devanagari)	0.0	100.0	16.0	+16.0
claude-haiku-4-5	Romanized Hindi	0.0	100.0	21.0	+21.0
claude-haiku-4-5	Hinglish	0.0	100.0	16.0	+16.0
llama-3.1-8b	English	0.3	99.5	—	—
llama-3.1-8b	Hindi (Devanagari)	1.0	98.5	22.5	+21.5
llama-3.1-8b	Romanized Hindi	0.0	100.0	23.0	+23.0
llama-3.1-8b	Hinglish	0.7	99.0	16.5	+15.8

Llama-3.3-70B, Claude Haiku and Llama-3.1-8B are almost deterministic (self-flip ≤1%), so almost all of their cross-language flips are caused by the language change. Gemma-4 and Qwen3 have a re-run noise floor of 7–11%. For Gemma on Hinglish, the excess is about 0, so most of its Hinglish flips there are noise.

Table 3. GEE logistic model (387 item clusters, n=18,576), odds ratio of a correct verdict vs English, adjusted for judge and order.

Language	OR	95% CI	p
Hindi (Devanagari)	0.747	[0.682, 0.818]	3.0e-10
Romanized Hindi	0.603	[0.542, 0.671]	1.8e-20
Hinglish	0.714	[0.653, 0.779]	6.1e-14

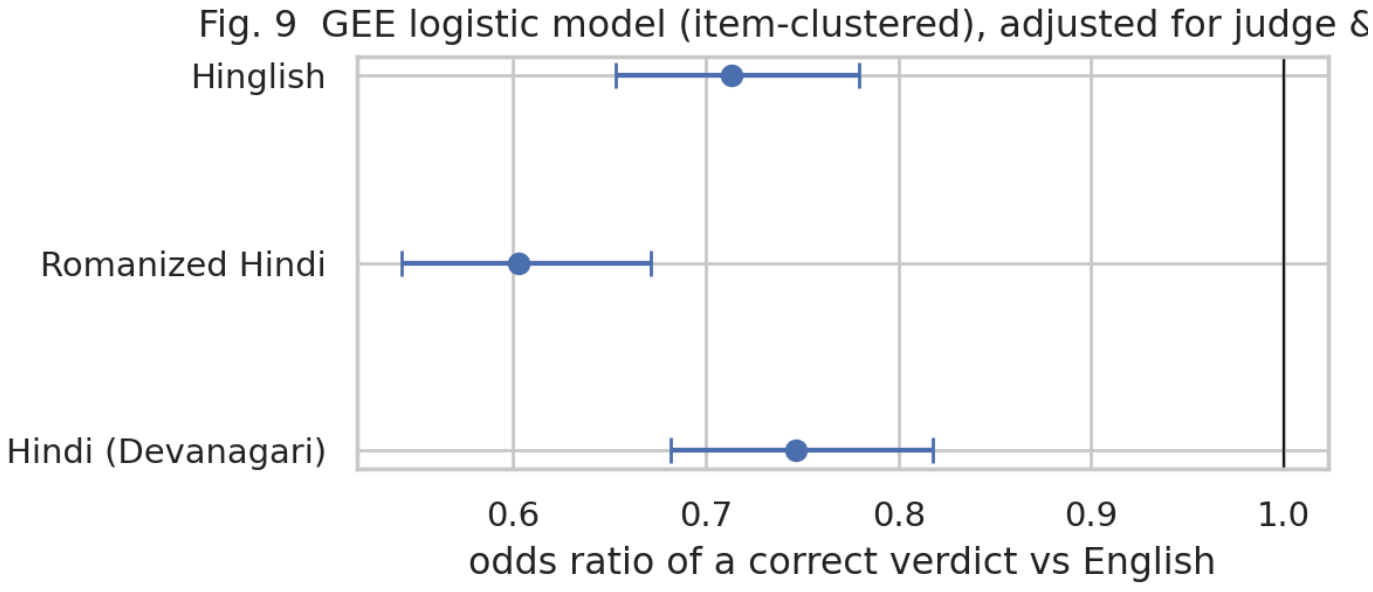


Figure 9. GEE odds ratios with 95% CIs.

Fig. 5 Position bias by judge and language

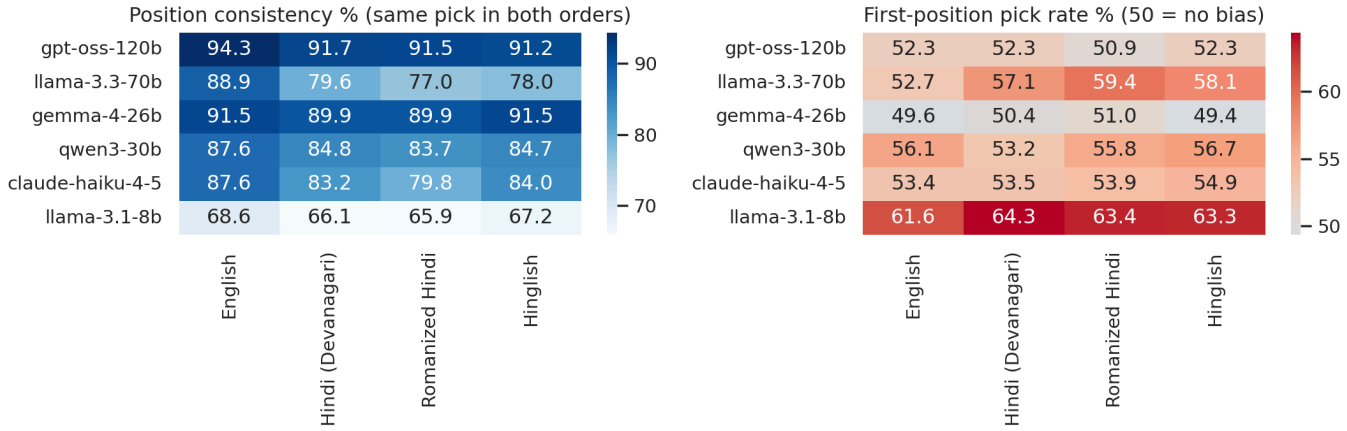


Figure 5. Position consistency and first-position preference. Switching away from English lowers consistency for most judges.

Fig. 6 Confusion matrices: gold better output vs judge pick (valid judgments)

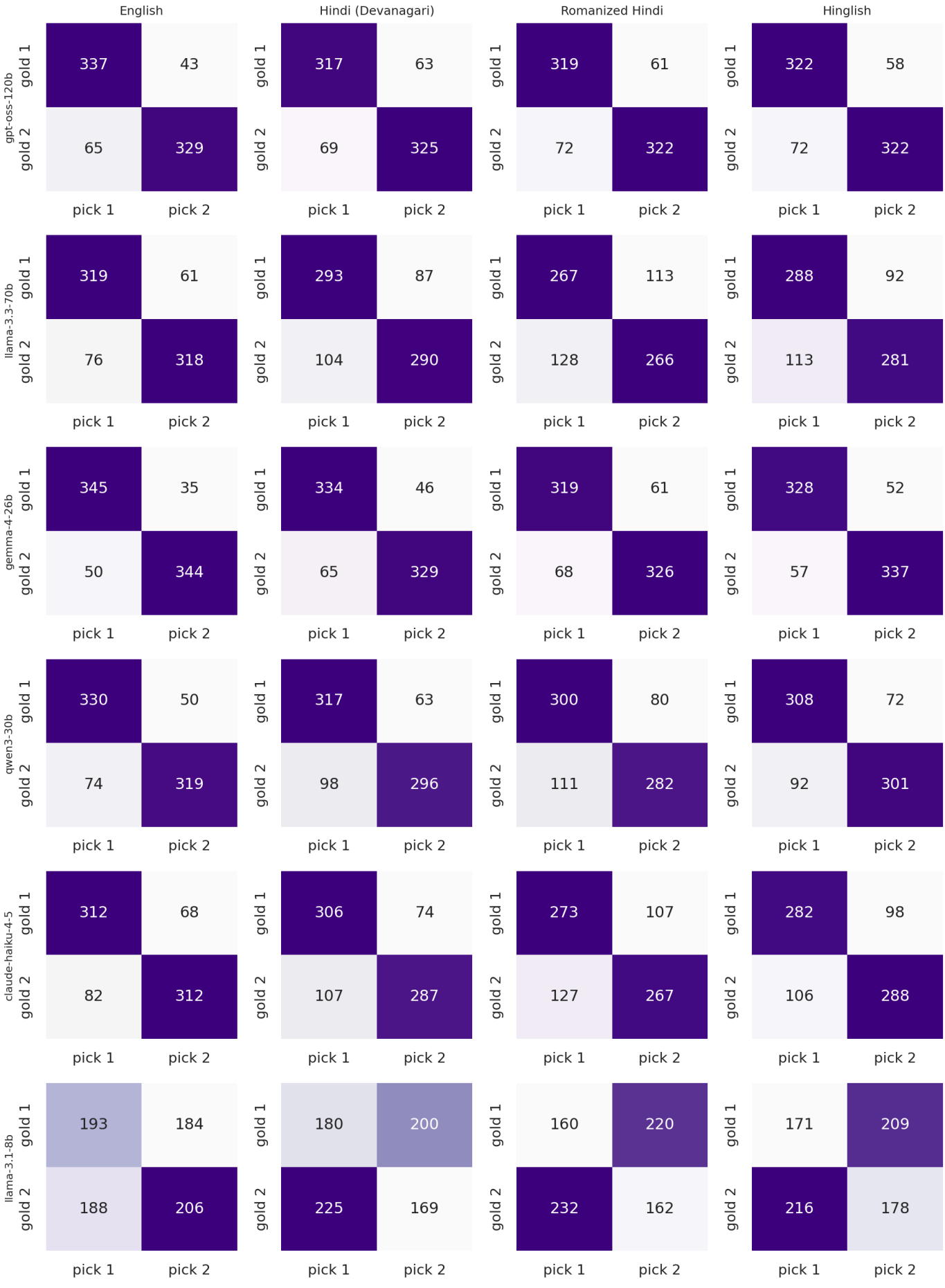
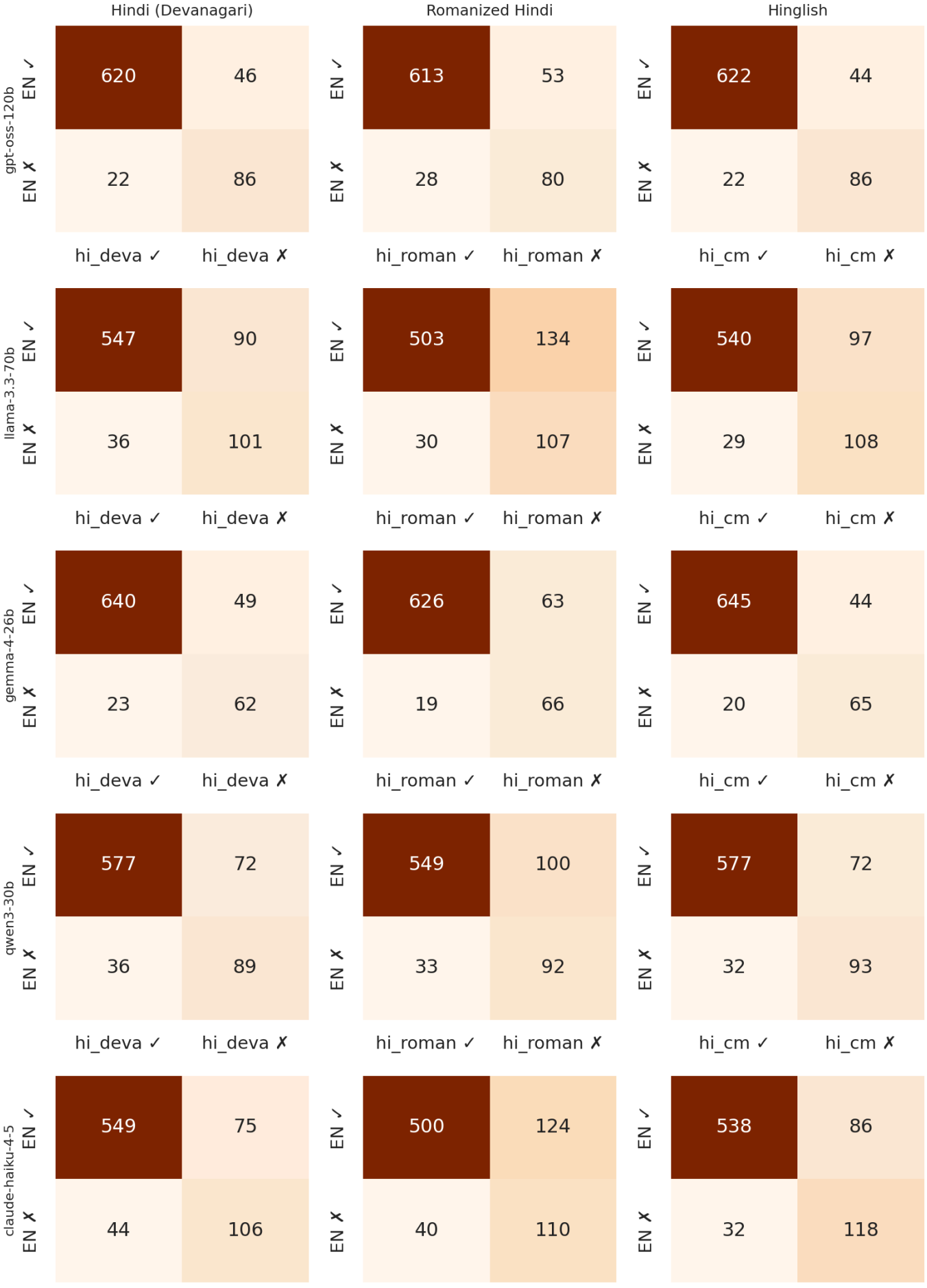


Figure 6. Confusion matrices (gold better output vs judge pick), valid judgments only. Llama-3.1-8B is near chance in English and anti-correlated with gold in Hindi conditions.

Fig. 7 Paired English-vs-variant correctness (McNemar tables)



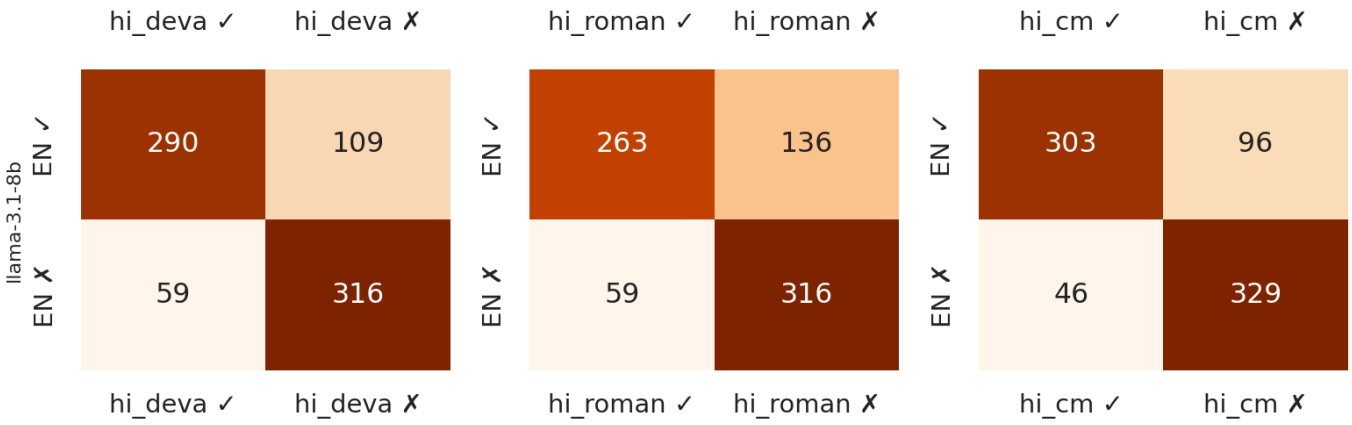


Figure 7. Paired correctness tables (English vs each language) used for the McNemar tests. The off-diagonal asymmetry (EN✓ → ✗ ≫ EN✗ → ✓) is the effect.

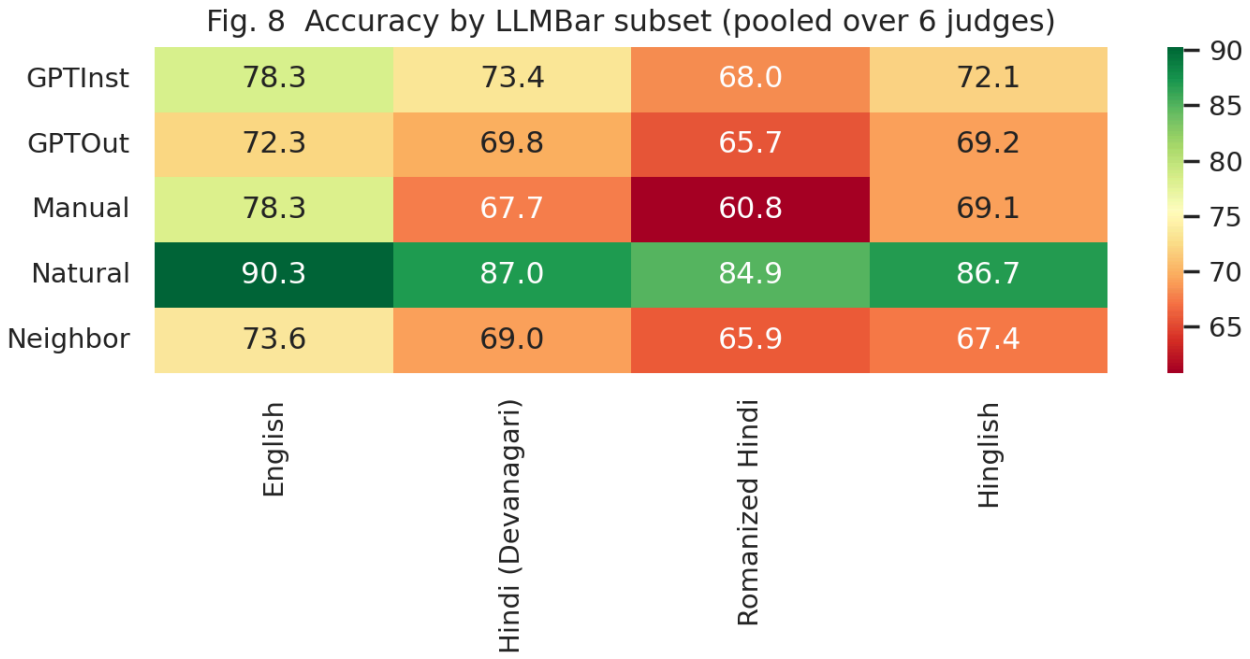


Figure 8. Accuracy by LLMBar subset, pooled over judges.

5. Robustness

Table 4. Mean Δacc (pp) across the six judges, and the number of judge × language cells significant after Holm correction.

Item set	Devanagari	Romanized	Hinglish	Significant
All 387 items	-4.8	-8.6	-5.6	18/18
Without 12 code-heavy items (375)	-4.8	-8.9	-5.6	18/18
Without 34 human-fixed items (353)	-4.4	-8.3	-4.8	18/18

The full per-judge tables for each subset are in metrics_no_codeheavy.csv and metrics_no_humanfixed.csv. Counting the 6 unreadable judgments as incorrect vs dropping them changes accuracy by at most 0.2 pp (Table 1 shows only the conservative version; both are in the CSV).

6. Limitations (to state in the paper)

- Translations were made by machine (Gemini 2.5 Pro), checked by Claude Sonnet 4.5, and human-validated by two raters on 251 cells (raw meaning agreement 91%, Gwet's AC1 0.91). A drop in accuracy may partly reflect residual translation quality rather than judge bias, although the effect holds after removing human-fixed items.
- One benchmark (LLMBar), one prompt template, and six judges accessed through Cloudflare/Anthropic APIs at a fixed date. Results are a measurement for these settings, not a universal claim.
- Claude Haiku 4.5 belongs to the same family as the verifier model (Claude Sonnet 4.5), not the translator.
- Cost is a token-based estimate; the exact billing is in the AI Gateway logs.

Code: src/judge.py, src/check_step4.py, src/step5_analysis.py, src/build_report.py. Raw data: results/step4/*.jsonl.